MANAN MITTAL

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Machine Learning Engineer with hands-on experience in developing intelligent systems, including recommendation engines, image captioning models, and working with advanced LLM pipelines such as GANs and RAG. Adept at applying deep learning and traditional machine learning methods to solve practical problems with clarity and precision. Strong analytical abilities combined with a deep interest in building AI solutions that bridge the gap between complex algorithms and real-world impact.

Skills

Machine Learning & Deep Learning:
Supervised & Unsupervised Learning Models, CNN, RNN, Recommendation Systems, Image Captioning, Generative Adversarial Networks (GANs), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs).
Programming Languages:
Python, C++, Java
Tools & Libraries:
NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Matplotlib, Seaborn
Platforms & Technologies:
IBM Watsonx (Gen AI) , Git, Google Colab, APIs for AI/ML, AWS (EC2)

Projects

- Developed an Image Captioning Model using ResNet50 for feature extraction and LSTM for sequence generation, achieving a BLEU-1 score of 0.514 and BLEU-4 score of 0.289 on standard datasets.
- Built IkigAI, a career recommendation system inspired by the Ikigai framework, using real-world data (200 samples, 15 features). Achieved strong results with LightGBM (F1: 0.84), Random Forest (F1: 0.81), and XGBoost (F1: 0.86).
- Created MythGenerator, a mini-project using the Mixtral-8x7B-Instruct model to generate creative stories based on user-selected country and theme, deployed via Streamlit at **mythgenerator.streamlit.app.**

Internships

AI Intern – IBM Generative AI Internship *May 2024 – July 2024*

- Gained hands-on experience working with **Generative AI technologies** through **IBM Watsonx** platform.
- Explored practical applications of LLMs, prompt engineering, and retrieval-augmented generation (RAG) pipelines.

Certificates

- **Matlab Onramp** (MathWorks)
- **Applied Machine Learning** (Coursera)
- **Java** (HackerRank) – Certified with **Gold Badge** for proficiency
- **GenAI using IBM Watsonx** (IBM)

Educational Background

- **Schooling**

DAV ACC Public School, Katni (MP) 10th - **94.33%** *2019-2020*
Diamond Eng. Hr. Sec. School, Katni (MP) 12th - **78.60%** *2021-2022*

- **Undergraduate Degree (Ongoing)**

VIT Bhopal University – B. Tech in CSE with specialization in AI & ML *2023-present*

References

Github - https://github.com/CypherManan
Leetcode - https://leetcode.com/u/CypherManan/
Kaggle - https://www.kaggle.com/cyphermanan
Linkedin - https://www.linkedin.com/in/manan-mittal-822477308/